DATA WAREHOUSE PROJECT DOCUMENTS

AHMET CAPAR

DATA WAREHOUSE – GENERAL PROJECT PLAN

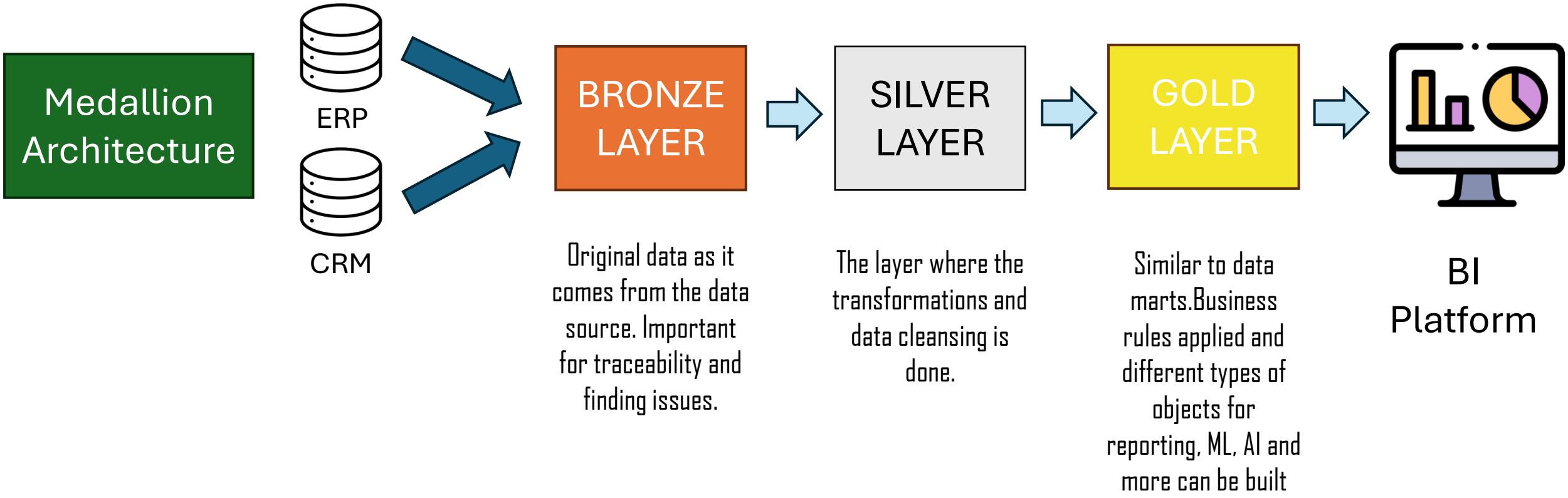
(Click on the arrows to see the related page or document)

General Plan for Data Warehouse Project		
Main phases	**Tasks**	**Action**
Requirement analysis	Analyse the requirements of the project based on the data sets	Data Warehouse Project Project Requirements document prepared.
	Choose data management approach	Medallion data architecture is chosen.
	Design of the layers	See layers design
	Draw data architecture	See data architecture
Project initialization	Build Bronze, Silver and Gold Layers	See detailed plan to build layers of the data warehouse.
	Define project naming conventions	See 'Naming Conventions for Data Warehouse Project' document.
	Create Git repo and prepare repo structure	See the Git repo via this link
	Create database and schemas	Database created. Schemas for Bronze, Sliver and Gold layers are created

DATA WAREHOUSE – DETAILED PROJECT PLAN

Detailed plan for building Bronze, Silver and Gold layers		
Main Phase	**Task**	**Action**
Bronze Layer	Analyse source systems	See 'Bronze Layer Tables' page
	Write the code for data ingestion. Validate data completeness and check schemas	1. See 'init_database.sql' that creates tables in bronze layer. 2. A procedure for loading the source data (in csv files) to bronze layer is created. See scripts/proc_load_bronze.sql The procedure checks and validates data completenes and checks schemas.
	Draw data flow for documentation	See data flow chart document
	Commit code in Git Repo	See scripts/bronze
Silver Layer	Explore and undertand data content	See Bronze Layer Tables and Integration Model pages
	Write the code for data cleansing	See the code in GitHub: https://github.com/kazurla/sql-server-data-warehouse-project-1/blob/main/scripts/silver/data_clean_and_transform_silver.sql
	Validate data correctness	All validations done in load procedure
	Documentation of the silver layer	See the Data Flow page in presentation
	Commit code in Git Repo	See the files in GitHub Repo
Gold Layer	Explore business requirements and objects	See page
	Write code for data integration	See code in github
	Validate data integration	All validations are done in load procedure
	Draw star schema data model for documentation	See page
	Create data catalog for documentation	See page
	Commit code in Git Repo	All documents, scripts for gold layer is uploaded.

DATA WAREHOUSE ARCHITECTURE



Back to the General Project Plan

DATA WAREHOUSE LAYERS

	BRONZE LAYER *« ingest »*	SILVER LAYER *« clean »*	GOLD LAYER *« business »*
Definition	Raw, unprocessed data, as it comes from the source	Clean, standardized data	Business-ready data
Objective	Tracablilty and debugging	Data preperation for analysis	Getting data ready for analytics and reporting
Object Type	Tables	Tables	Vievs
Load Method	Full Load (Truncate and insert)	Full Load (Truncate and insert)	None
Data Transformation	None (as-is)	• Data cleaning • Data Standardization • Data normalization • Data enrichment • Derived Columns	• Data integration • Data aggregation • Business logic and values
Data Modelling	None (as-is)	None (as-is)	Star schema / Aggregated objects / Flat tables
Target Audience	Data engineers	• Data engineers • Data analysts	• Data analysts • Business users

Back to the General Project Plan

HIGH LEVEL ARCHITECTURE

Sources

CRM

ERP

Object Type: CSV FILES
Interface : FILES IN FOLDERS

Data Warehouse

Microsoft SQL Server

BRONZE LAYER
« ingest »

Raw Data

Object Type: TABLES

Load:
- Batch processing
- Full load
- Truncate & Insert

No transformations

Data Model: None (as-is)

SILVER LAYER
« clean »

Cleaned,standardized Data

Object Type: TABLES

Load:
- Batch processing
- Full load
- Truncate & Insert

Transformations
- Data cleaning
- Data Standardization
- Data normalization
- Data enrichment
- Derived Columns

Data Model: None (as-is)

GOLD LAYER
« business »

Business – ready Data

Object Type: VIEWS

Load: No load

Transformations
- Data integration
- Data aggregation
- Business logic and values

Data Model: None (as-is)

Consume

BI & Reporting

Ad-Hoc SQL queries

ML / DL etc...

Back to the General Project Plan

STEPS TO BUILD THE BRONZE LAYER

BRONZE LAYER

Analysing
Interview source system experts

Coding
Data Ingestion

Validating
Data Completeness and schema checks

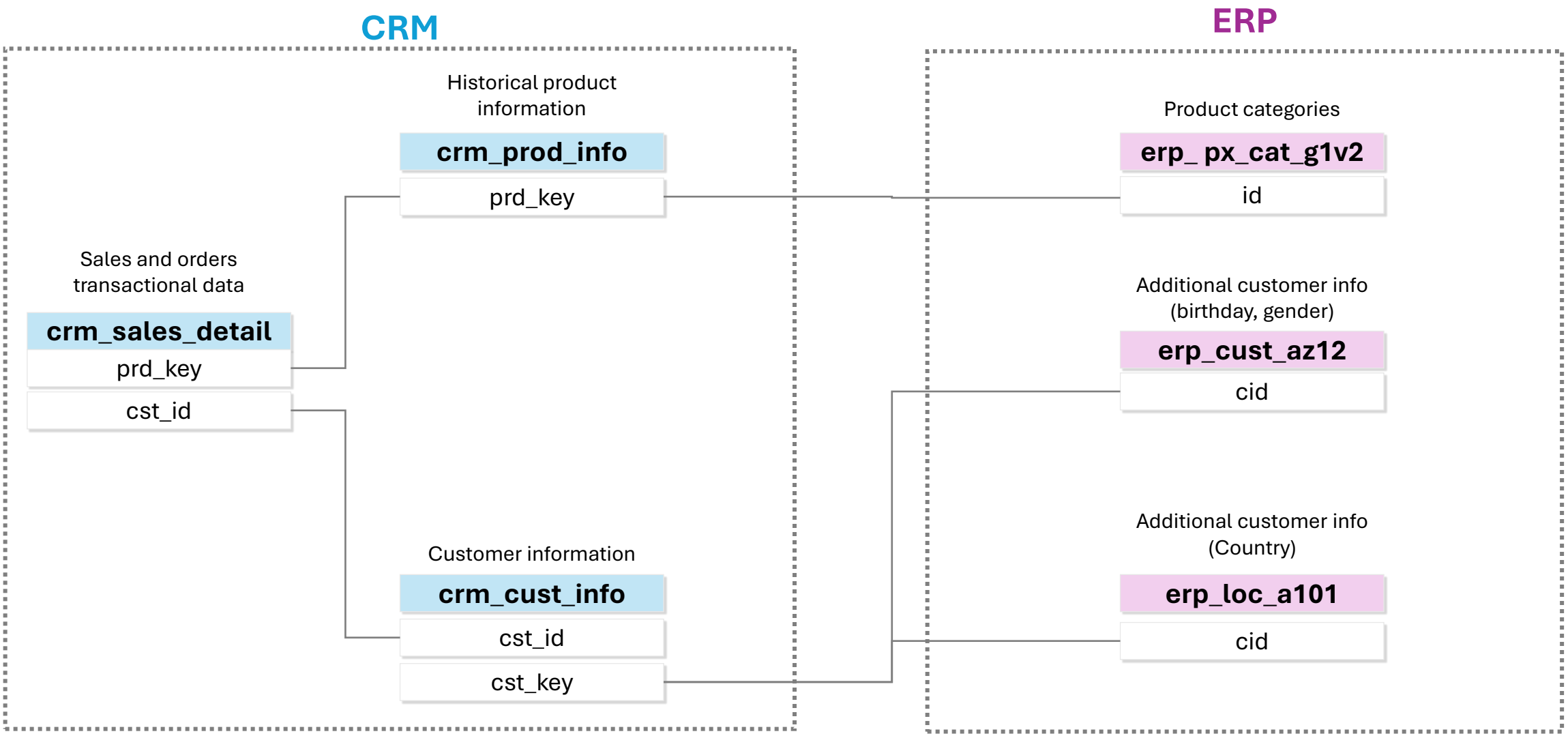
Docs and Versions
Documenting, versioning in Git

DATA WAREHOUSE BRONZE LAYER

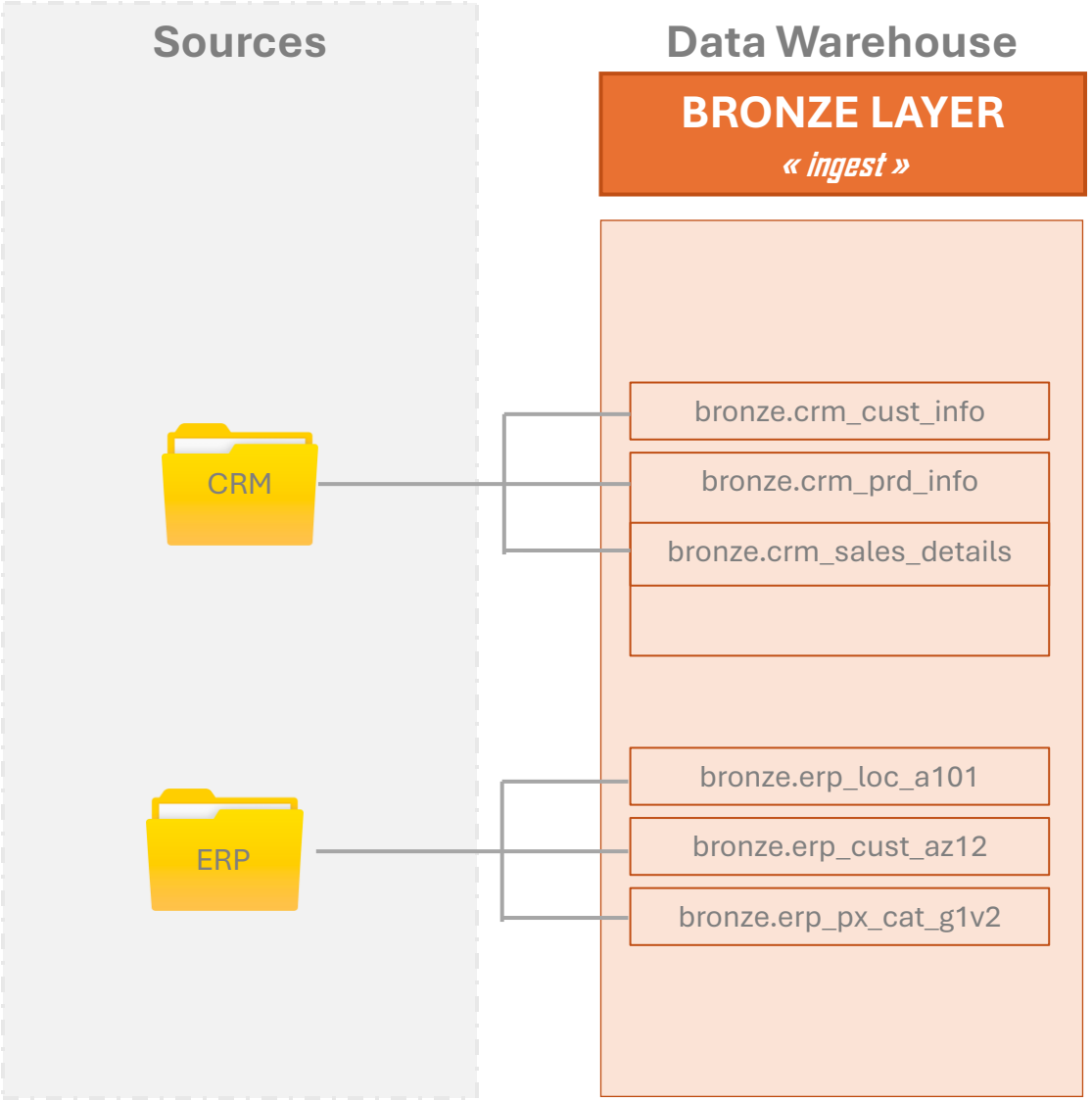
	BRONZE LAYER *« ingest »*
Definition	Raw, unprocessed data, as it comes from the source
Objective	Tracablilty and debugging
Object Type	Tables
Load Method	Full Load (Truncate and insert)
Data Transformation	None (as-is)
Data Modelling	None (as-is)
Target Audience	Data engineers

BRONZE LAYER – INTEGRATION MODEL

(How the tables are connected?)



DATA FLOW (SOURCE TO BRONZE LAYER)



BRONZE LAYER - TABLES

CRM

bronze_cust_info	
cst_id	INT
cst_key	NVARCHAR(50)
cst_firstname	NVARCHAR(50)
cst_lastname	NVARCHAR(50)
cst_marital_status	NVARCHAR(50)
cst_gndr	NVARCHAR(50)
cst_create_date	DATE

bronze_prd_info	
prd_id	INT
prd_key	NVARCHAR(50)
prd_nm	NVARCHAR(50)
prd_cost	INT
prd_line	NVARCHAR(50)
prd_start_dt	DATETIME
prd_end_dt	DATETIME

bronze_sales_details	
sls_ord_num	NVARCHAR(50)
sls_prd_key	NVARCHAR(50)
sls_cust_id	INT
sls_order_dt	INT
sls_ship_dt	INT
sls_due_dt	INT
sls_sales	INT
sls_quantity	INT
sls_price	INT

ERP

bronze_loc_a101	
cid	NVARCHAR(50)
cntry	NVARCHAR(50)

bronze_cust_az12	
cid	NVARCHAR(50)
bdate	DATE
gen	NVARCHAR(50)

bronze_px_cat_g1v2	
id	NVARCHAR(50)
cat	NVARCHAR(50)
subcat	NVARCHAR(50)
maintenance	NVARCHAR(50)

STEPS TO BUILD THE SILVER LAYER

BRONZE LAYER

Analysing
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Coding
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SILVER LAYER

Analysing
Explore and understand data

Coding
Data Cleansing

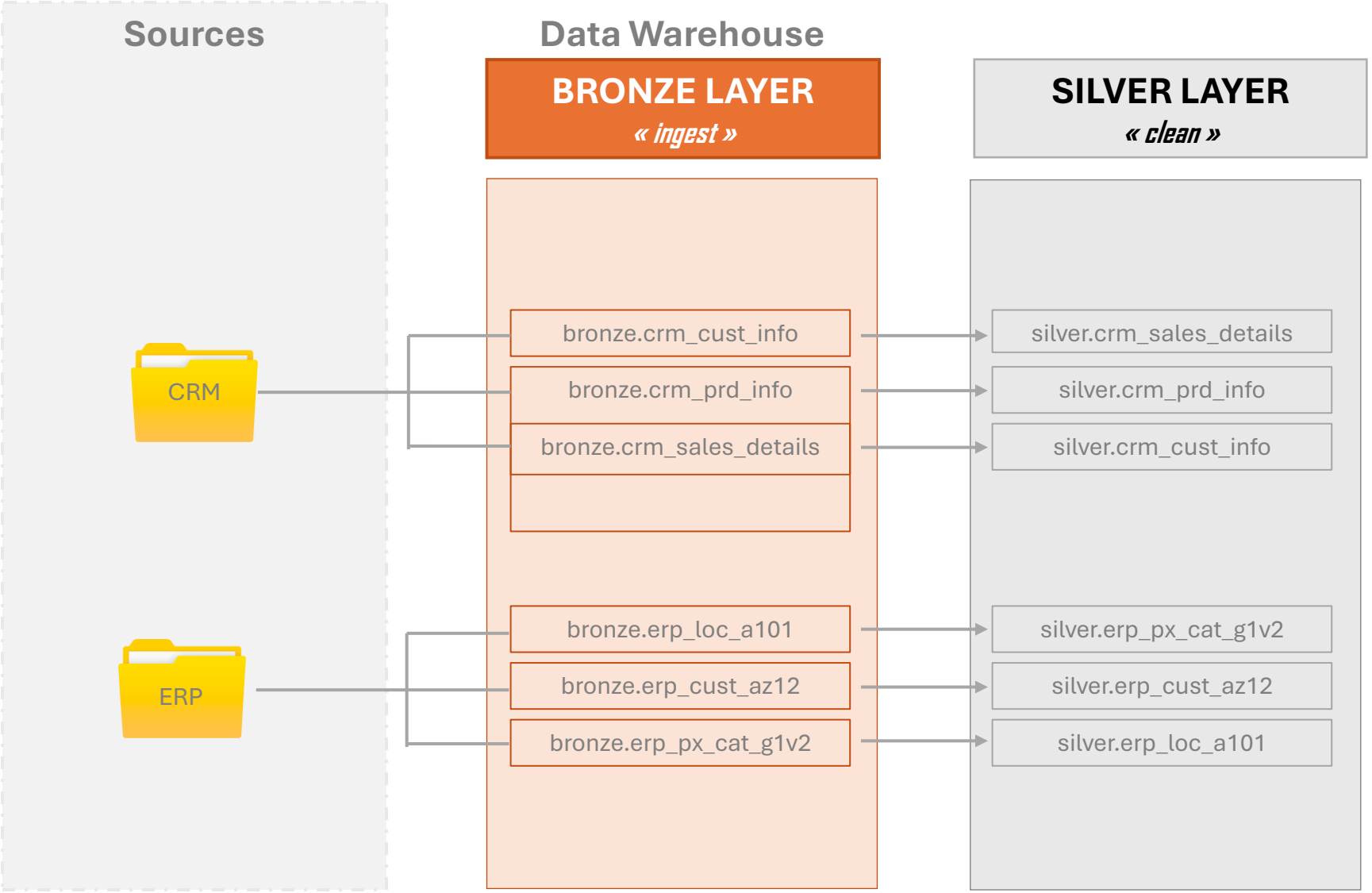
Validating
Data completeness and correction checks

Docs and Versions
Documenting, versioning in Git

DATA WAREHOUSE SILVER LAYER

	SILVER LAYER *« clean »*
Definition	Clean, standardized data
Objective	Data preperation for analysis
Object Type	Tables
Load Method	Full Load (Truncate and insert)
Data Transformation	• Data cleaning • Data Standardization • Data normalization • Data enrichment • Derived Columns
Data Modelling	None (as-is)
Target Audience	• Data engineers • Data analysts

DATA FLOW (BRONZE LAYER TO SILVER LAYER)



SILVER LAYER – DATA CLEANSING AND TRANSFORMATION STEPS

1. Check and Remove Duplicates or nulls in the primary keys
2. Check unwanted spaces in string values (cst_firstname, cst_lastname)
3. Data Standardisation and Handling Missing Data
4. Derive new columns to match the formats of data
5. Check the quality of data and handle missing data
6. Data Enrichment and data type casting
7. Transforming an integer to date

To see some examples,
Click on the arrow

GOLD LAYER – BUSINESS OBJECTS

FACT TABLES

SALES

crm_sales_detail

prd_key

cst_id

Sales and orders transactional data

DIMENTION TABLES

PRODUCT

crm_prod_info

prd_key

Historical product information

erp_ px_cat_g1v2

id

Product categories

Join product related tables to create the product dimention

CUSTOMER

crm_cust_info

cst_id

cst_key

Customer information

erp_cust_az12

cid

Additional customer info (birthday, gender)

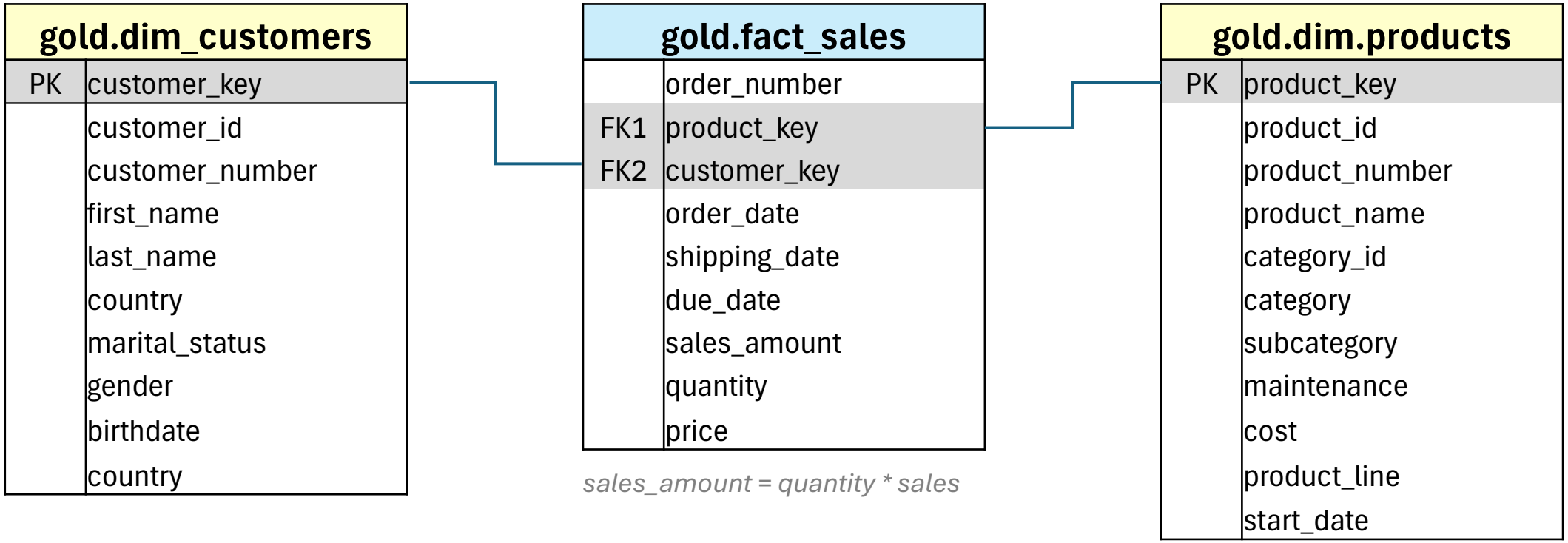
erp_loc_a101

cid

Additional customer info (Country)

Join customer related tables to create the customer dimention

GOLD LAYER - INTEGRATION



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GOLD LAYER

Analysing
Explore and understand Business Objects

Coding
Data Integration

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DATA WAREHOUSE LAYERS

	GOLD LAYER *« business »*
Definition	Business-ready data
Objective	Getting data ready for analytics and reporting
Object Type	Vievs
Load Method	None
Data Transformation	• Data integration • Data aggregation • Business logic and values
Data Modelling	Star schema / Aggregated objects / Flat tables
Target Audience	• Data analysts • Business users